

Objective: To work in an environment offering challenging career opportunities where I can apply and enhance my skills for solving diverse technical problems in Data Science, Machine Learning, and AI Systems.

TECHNICAL SKILLS

Languages: Python, Java, SQL
AI & Machine Learning: AutoML, XGBoost, LightGBM, CatBoost, Scikit-learn, RAG, ChromaDB, Ollama, Predictive Modeling, Regression, Classification, Feature Engineering, Hyperparameter Tuning, Benchmarking, EDA
NLP & Speech AI: Speech Recognition (SR), Natural Language Processing (NLP), Text-to-Speech (TTS), Sentence Transformers, BM25 Hybrid Search
Frameworks & Tools: FastAPI, React.js, Streamlit, Docker, Python Data Pipelines

EXPERIENCE

Data Scientist Intern — Gradtwin (Chennai, Tamil Nadu) June 2026 – August 2026

- Architected end-to-end Python data pipelines and performed rigorous EDA on multi-source datasets to uncover actionable business insights.
- Developed predictive modeling workflows (regression & classification), improving forecasting via feature engineering and hyperparameter tuning.
- Designed & deployed a production-grade RAG AI chatbot integrating LLM APIs, ChromaDB vector storage, and semantic document retrieval.
- Cleaned and engineered features from complex, high-volume datasets; collaborated cross-functionally to implement ML-driven solutions.

FEATURED PROJECTS

AI-Powered Data Cleaning and Imputation Engine (DataClean AI) Aug 2026

Stack: FastAPI, React.js, Scikit-learn, XGBoost, LightGBM, CatBoost, Docker.

- Autonomously detects 13+ data anomalies (MCAR/MAR missingness, multivariate outliers, distribution skewness, class imbalance).
- Combines 30+ domain heuristic rules with a Voting Ensemble classifier achieving 97.8% recommendation accuracy across 25 techniques.
- Implements a 10MB chunked upload pipeline handling multi-gigabyte datasets (3M+ rows) with zero-OOM memory streaming and PDF reporting.

PDF Summarizer Chatbot using RAG (PaperBrain) Jul 2026

Stack: FastAPI, Streamlit, ChromaDB, Sentence Transformers, BM25, Ollama (Local LLMs).

- Fully offline, privacy-first RAG system where embeddings, vector storage, and language generation run locally without data leaving machine.
- Fuses dense vector similarity search (Sentence Transformers) with BM25 keyword matching for context retrieval and QA.

The Voice Assistant with Desktop Accessibility Mar 2026

Stack: Python, Speech Recognition (SR), NLP Intent Engine, Text-to-Speech (TTS), OS Automation.

- Hands-free desktop voice assistant with Speech Recognition, NLP command parsing, and responsive TTS audio feedback.

Residential Property Valuation Using Machine Learning Feb 2025

Stack: Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn.

- Predictive regression modeling on real estate datasets to accurately forecast property market valuations based on key features.

EDUCATION

BE - Computer Science and Engineering (Vinayaka Mission's KVEC, Salem • 2022–2026) 73.76%

Higher Secondary School Examination (HSC) (Govt Higher Secondary School, Salem • 2021–2022) 52.8%

Secondary School Examination (SSLC) (Govt Higher Secondary School, Salem • 2019–2020) 72.5%

CERTIFICATES & ACHIEVEMENTS

- GenAI Powered Data Analytics Job Simulation (Jan 2026)** — Completed Tata (via Forage) simulation in EDA & AI risk profiling. [ID: [SFPHHSaWj4nHTiKK7](#)]
- 1st Prize Winner – Kongu Engineering College IPL AUCTION 2025** — Analytics, valuation, and bidding strategy optimization.
- Data Science in Python Training Program (2024)** — IIIT KOTTAYAM Engineering College intensive training program.
- Green Skill and AI Course Certification (2025)** — Edunet Foundation course on AI fundamentals & sustainability.
- Paper Presentation on 'IOT in Military Security' (2023)** — KSR Engineering College technical presentation.